Program : **Diploma in Wood and Paper Technology**	
Course Code : **5123A**	Course Title: **Statistical Quality Assurance**
Semester : **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4, T:0, P:0)**	Periods per semester: **60**

Course Objectives:

- To impart knowledge on Quality, Quality Control And Assurance
- To introduce the concept of Statistical Quality Control.
- To illustrate the Control Chart.
- To design acceptance Sampling and O C Curve.

Course-prerequisites:

Topic/Description	Course code	Course Title	Semester
Knowledge of basic Mathematics		Mathematics I	1
Knowledge in Statistics		Mathematics II	2

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Outline the Quality, Quality Control And Assurance	15	Understanding
CO2	Analyze the Statistical quality control	14	Applying
CO3	Prepare the control chart	15	Applying
CO4	Solve acceptance sampling and o c curve.	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1							
CO2	3						
CO3			3				
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Outline the Quality, Quality Control And Assurance**		
M1.01	Explain quality	3	Remembering
M1.02	Discuss quality assurance system	4	Remembering
M1.03	Explain quality assurance program.	4	Understanding
M1.04	Summarize quality in materials management	4	Understanding
Contents: Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management.			
CO2	**Analyze the Statistical quality control**		
M 2.01	Explain statistical quality chart	3	Understanding
M 2.02	Develop frequency program.	4	Applying
M 2.03	Calculate mean ,mode and median	4	Understanding
M 2.04	Apply regression analysis	3	Applying
	Series Test – I	1	

Contents:

Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspectionand sampling Inspection.

CO3	**Prepare the control chart**		
M3.01	Explain variable control chart	5	Understanding
M3.02	Develop different chart in SQC	5	Applying
M3.03	Develop solution based on P chart/ U Chart	5	Applying

Contents:

Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart

CO4	**Solve acceptance sampling and o c curve.**		
M4.01	Explain sampling	3	Remembering
M4.02	Develop the difference between sampling and control chart	4	Applying
M4.03	Distinguish single sampling and double sampling plan	3	Applying
M4.04	Apply the problems on reliability	4	Applying
	Series Test – II	1	

Contents:

Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts – Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems

Text /Reference:

T/R	Book Title/Author
T1	M.Mahajan. Published by Dhanpatharai & company
R2	Industrial Engg and management by O.P. Khanna
R3	M.Mahajan. Statistical Quality control ,S Chand
R4	K.C.Aroa,.Total Quality Management ,Dhanapat

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=CfCztL5DkTE
2	https://www.youtube.com/watch?v=qb3mvJ1gb9g
3	https://www.youtube.com/watch?v=QsLDBfApqoI
4	https://www.youtube.com/watch?v=e5g2NmIUdck
5	https://www.youtube.com/watch?v=GC6k_qbpwmA